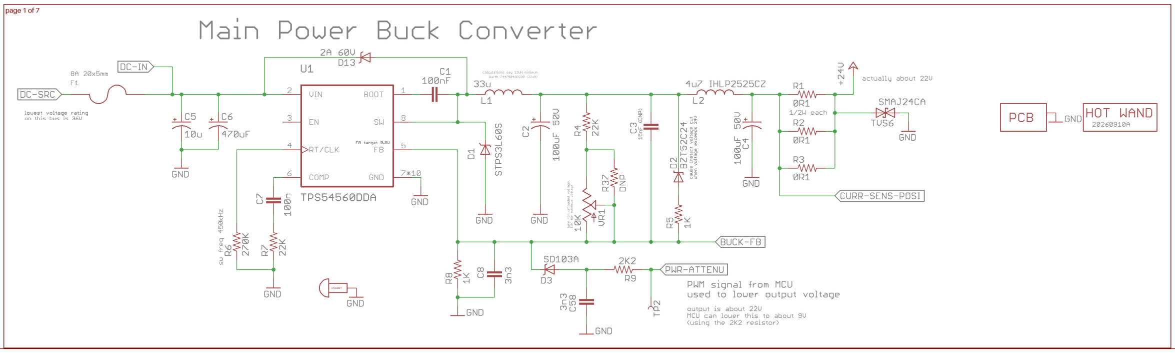
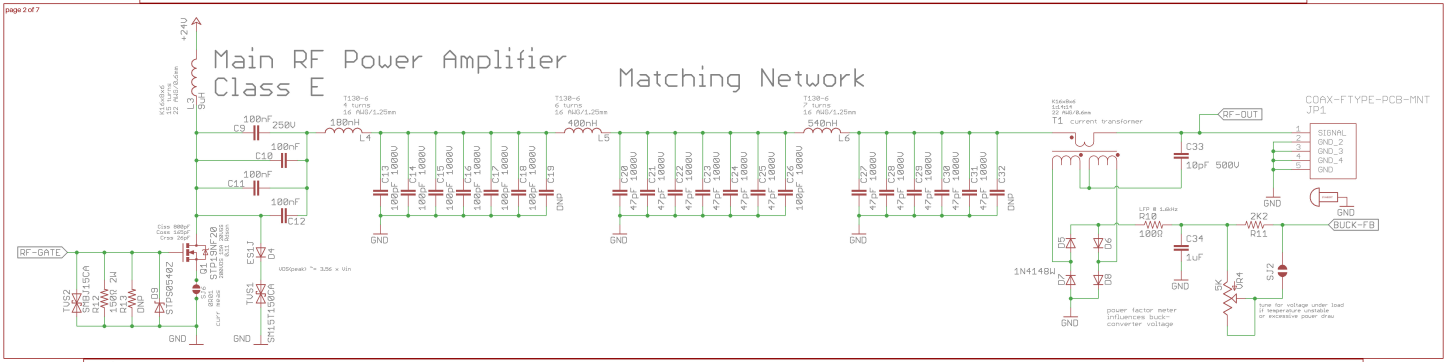
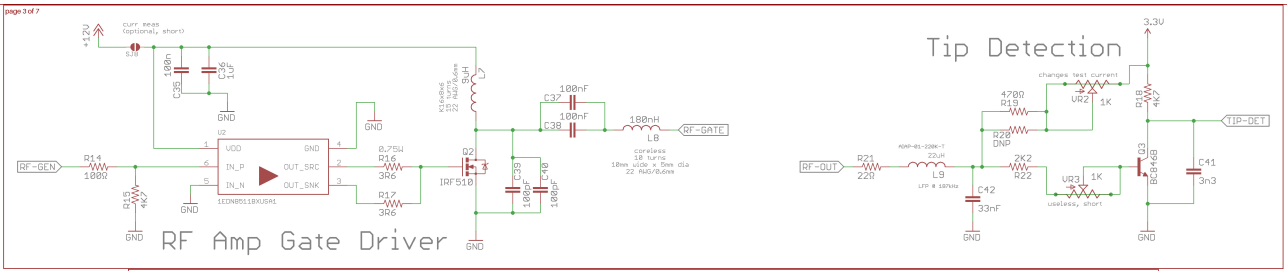
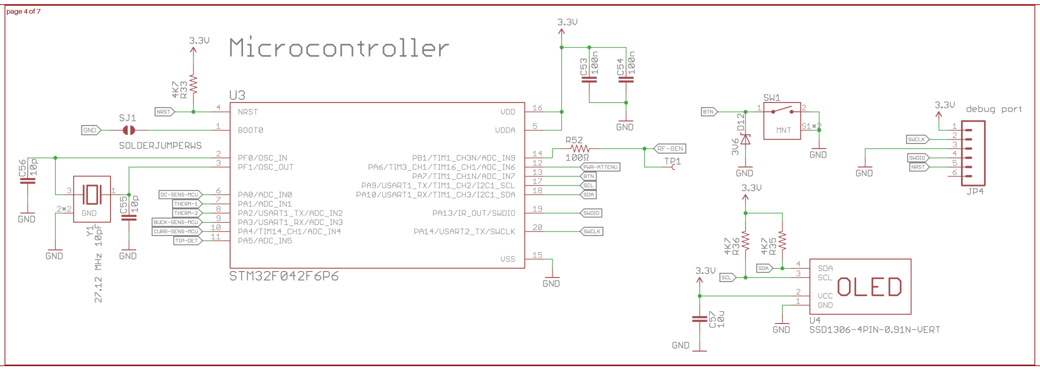
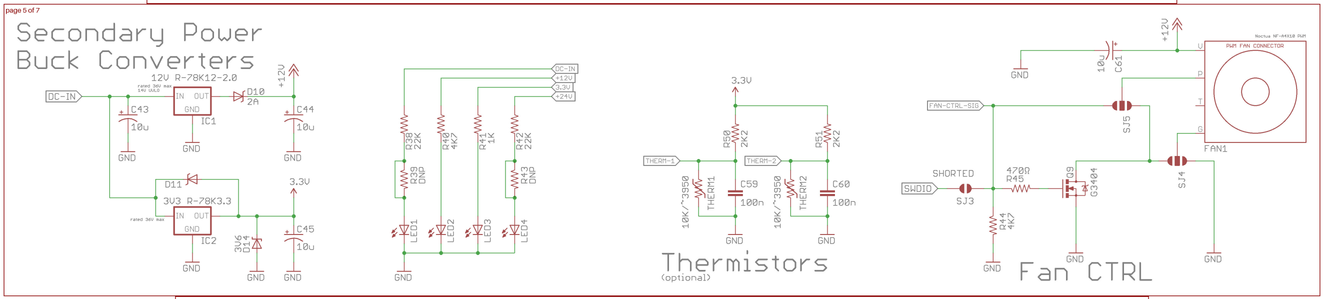
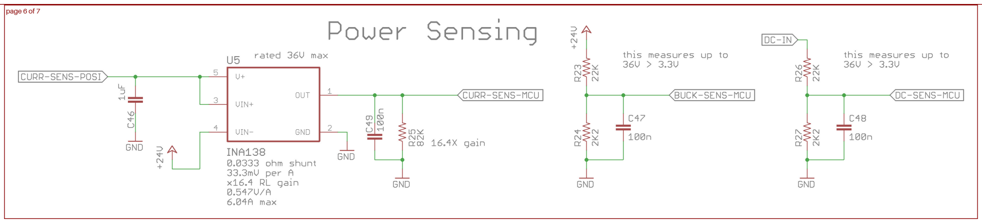

page 1 of 7
Main Power Buck Converter
DC-SRC
8A 20x5mm
F1
lowest voltage rating
on this bus is 36V
DC-IN
C5
10u
C6
470uF
GND
2A 60V
D13
U1
VIN
EN
RT/CLK
COMP
BOOT
SW
FB target 0.8V
FB
GND
7*10
TPS54560DDA
C7
100n
sw freq 450kHz
R6
270K
R7
22K
C1
100nF
calculations say 13uH minimum
Wurth 744750460220 (22uH)
33u
L1
D1
STPS3L60S
C2
100uF 50V
R4
22K
R37
DNP
tune for unloaded voltage
10K for maximum voltage
10K
VR1
C3
15nF (DNP)
D2
BZT52C24
cause instant voltage cut
when voltage exceeds 24V
R5
1K
4u7 IHLP2525CZ
L2
100uF 50V
C4
R1
0R1
1/2W each
R2
0R1
R3
0R1
+24V
actually about 22V
SMAJ24CA
TVS6
CURR-SENS-POSI
BUCK-FB
R8
1K
C8
3n3
SD103A
D3
2K2
R9
PWR-ATTENU
3n3
C58
TP2
PWM signal from MCU
used to lower output voltage
output is about 22V
MCU can lower this to about 9V
(using the 2K2 resistor)
PCB
HOT WAND
20260910A

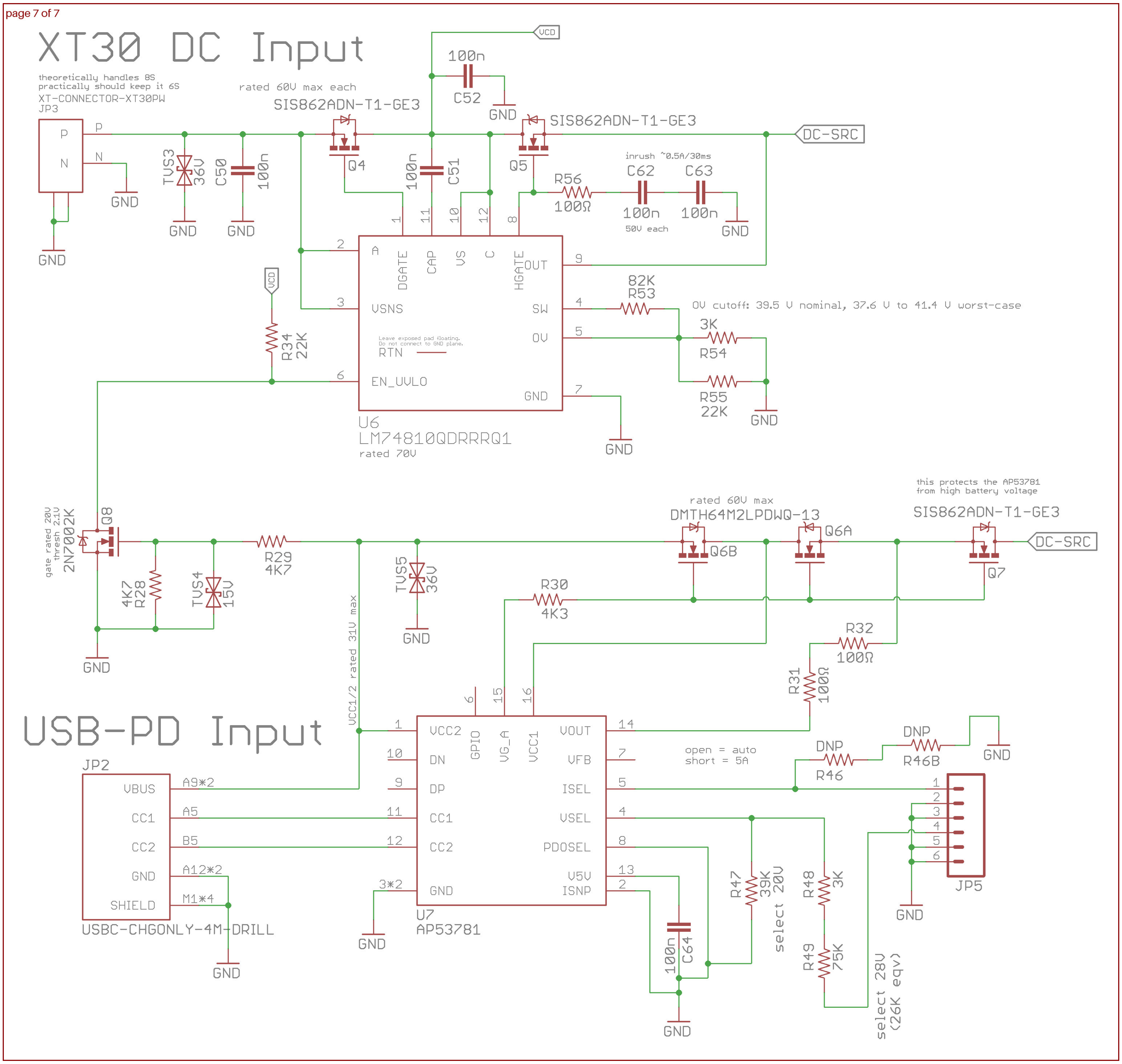

page 2 of 7
Main RF Power Amplifier
Class E
Matching Network
+24V
K16x8x6
15 turns
22 AWG/0.6mm
L3
9uH
C9
100nF 250V
C10
100nF
C11
100nF
C12
100nF
T130-6
4 turns
16 AWG/1.25mm
180nH
L4
C13 1000V 100pF
C14 1000V 100pF
C15 1000V 100pF
C16 1000V 100pF
C17 1000V 100pF
C18 1000V 100pF
C19 DNP
T130-6
6 turns
16 AWG/1.25mm
400nH
L5
C20 1000V 47pF
C21 1000V 47pF
C22 1000V 47pF
C23 1000V 47pF
C24 1000V 47pF
C25 1000V 47pF
C26 1000V 100pF
T130-6
7 turns
16 AWG/1.25mm
540nH
L6
C27 1000V 47pF
C28 1000V 47pF
C29 1000V 47pF
C30 1000V 47pF
C31 1000V 47pF
C32 DNP
K16x8x6
1:14:14
22 AWG/0.6mm
T1 current transformer
RF-OUT
C33
10pF 500V
COAX-FTYPE-PCB-MNT
JP1
SIGNAL
GND_2
GND_3
GND_4
GND
RF-GATE
TVS2
SMBJ15CA
R12
150Ω 2W
R13
DNP
D9
STPS0540Z
Ciss 800pF
Coss 165pF
Crss 26pF
Q1
STP19NF20
200VDS 15A 20VGS
0.11 Rdson
SJ6
0R01
curr meas
ES1J
D4
VDS(peak) ~= 3.56 x Vin
TVS1
SM15T150CA
GND
D5
D6
D7
D8
1N4148W
LFP @ 1.6kHz
R10
100Ω
C34
1uF
2K2
R11
BUCK-FB
5K
VR4
SJ2
power factor meter
influences buck-
converter voltage
tune for voltage under load
if temperature unstable
or excessive power draw


page 3 of 7
+12V
curr meas
(optional, short)
SJ8
100n
C35
C36
1uF
GND
U2
VDD
IN_P
IN_N
GND
OUT_SRC
OUT_SNK
1EDN8511BXUSA1
RF-GEN
R14
100Ω
R15
4K7
0.75W
R16
3R6
R17
3R6
Q2
IRF510
K16x8x6
15 turns
22 AWG/0.6mm
9uH
L7
C37
100nF
C38
100nF
C39
100pF
C40
100pF
180nH
L8
coreless
10 turns
10mm wide x 5mm dia
22 AWG/0.6mm
RF-GATE
RF Amp Gate Driver
Tip Detection
3.3V
changes test current
470Ω
R19
VR2
1K
R18
4K7
R20
DNP
TIP-DET
RF-OUT
R21
22Ω
AIAP-01-220K-T
22uH
L9
LFP @ 107kHz
C42
33nF
2K2
R22
VR3
1K
useless, short
Q3
BC846B
C41
3n3
GND


page 4 of 7
Microcontroller
3.3V
4K7
R33
NRST
U3
SJ1
GND
SOLDERJUMPERWS
C56
10p
Y1
2*2
GND
27.12 MHz 10pF
C55
10p
DC-SENS-MCU
THERM-1
THERM-2
BUCK-SENS-MCU
CURR-SENS-MCU
TIP-DET
NRST
BOOT0
PF0/OSC_IN
PF1/OSC_OUT
PA0/ADC_IN0
PA1/ADC_IN1
PA2/USART1_TX/ADC_IN2
PA3/USART1_RX/ADC_IN3
PA4/TIM14_CH1/ADC_IN4
PA5/ADC_IN5
VDD
VDDA
PB1/TIM1_CH3N/ADC_IN9
PA6/TIM3_CH1/TIM16_CH1/ADC_IN6
PA7/TIM1_CH1N/ADC_IN7
PA9/USART1_TX/TIM1_CH2/I2C1_SCL
PA10/USART1_RX/TIM1_CH3/I2C1_SDA
PA13/IR_OUT/SWDIO
PA14/USART2_TX/SWCLK
VSS
STM32F042F6P6
C53
100n
C54
100n
R52
100Ω
RF-GEN
TP1
PWR-ATTENU
BTN
SCL
SDA
SWDIO
SWCLK
SW1
MNT
S1*2
D12
3V6
debug port
JP4
4K7
R36
4K7
R35
SDA
SCL
VCC
GND
OLED
C57
10u
U4
SSD1306-4PIN-0.91N-VERT


page 5 of 7
Secondary Power
Buck Converters
12V R-78K12-2.0
rated 36V max
14V UVLO
DC-IN
C43
10u
IN
OUT
GND
IC1
D10
2A
+12V
C44
10u
D11
3V3 R-78K3.3
rated 36V max
IC2
3V6
D14
3.3V
C45
10u
GND
DC-IN
+12V
3.3V
+24V
R38
22K
R39
DNP
R40
4K7
R41
1K
R42
22K
R43
DNP
LED1
LED2
LED3
LED4
THERM-1
THERM-2
R50
2K2
R51
2K2
10K/~3950
THERM1
THERM2
C59
100n
C60
100n
Thermistors
(optional)
+12V
10u
C61
Noctua NF-A4X10 PWM
PWM FAN CONNECTOR
FAN1
FAN-CTRL-SIG
SJ5
SHORTED
SWDIO
SJ3
470Ω
R45
R44
4K7
Q9
G3404
SJ4
Fan CTRL


page 6 of 7
Power Sensing
U5
rated 36V max
CURR-SENS-POSI
1uF
C46
V+
VIN+
VIN-
OUT
GND
+24V
INA138
0.0333 ohm shunt
33.3mV per A
x16.4 RL gain
0.547V/A
6.04A max
C49
100n
R25
82K
16.4X gain
CURR-SENS-MCU
+24V
R23
22K
this measures up to
36V > 3.3V
BUCK-SENS-MCU
R24
2K2
C47
100n
DC-IN
R26
22K
this measures up to
36V > 3.3V
DC-SENS-MCU
R27
2K2
C48
100n
GND


XT30 DC Input
theoretically handles 8S
practically should keep it 6S
XT-CONNECTOR-XT30PW
JP3
P
N
P
N
GND
TVS3
36V
GND
C50
100n
GND
rated 60V max each
SIS862ADN-T1-GE3
Q4
VCD
100n
C52
GND
100n
C51
SIS862ADN-T1-GE3
Q5
DC-SRC
inrush ~0.5A/30ms
R56
100Ω
C62
100n
C63
100n
50V each
GND
A
DGATE
CAP
VS
C
HGATE
OUT
VSNS
SW
OV
Leave exposed pad floating.
Do not connect to GND plane.
RTN
EN_UVLO
GND
U6
LM74810QDRRRQ1
rated 70V
GND
82K
R53
OV cutoff: 39.5 V nominal, 37.6 V to 41.4 V worst-case
3K
R54
R55
22K
GND
VCD
R34
22K
gate rated 20V
thresh 2.1V
2N7002K
Q8
4K7
R28
TVS4
15V
R29
4K7
GND
TVS5
36V
GND
rated 60V max
DMTH64M2LPDWQ-13
Q6B
Q6A
this protects the AP53781
from high battery voltage
SIS862ADN-T1-GE3
Q7
DC-SRC
R30
4K3
R32
100Ω
R31
100Ω
VCC1/2 rated 31V max
USB-PD Input
JP2
VBUS
CC1
CC2
GND
SHIELD
A9*2
A5
B5
A12*2
M1*4
USBC-CHGONLY-4M-DRILL
GND
VCC2
DN
DP
CC1
CC2
GND
GPIO
VG_A
VCC1
VOUT
VFB
ISEL
VSEL
PDOSEL
V5V
ISNP
3*2
U7
AP53781
GND
open = auto
short = 5A
DNP
R46
DNP
R46B
GND
JP5
GND
R47
39K
select 20V
R48
3K
R49
75K
select 28V
(26K eqv)
100n
C64
GND
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16